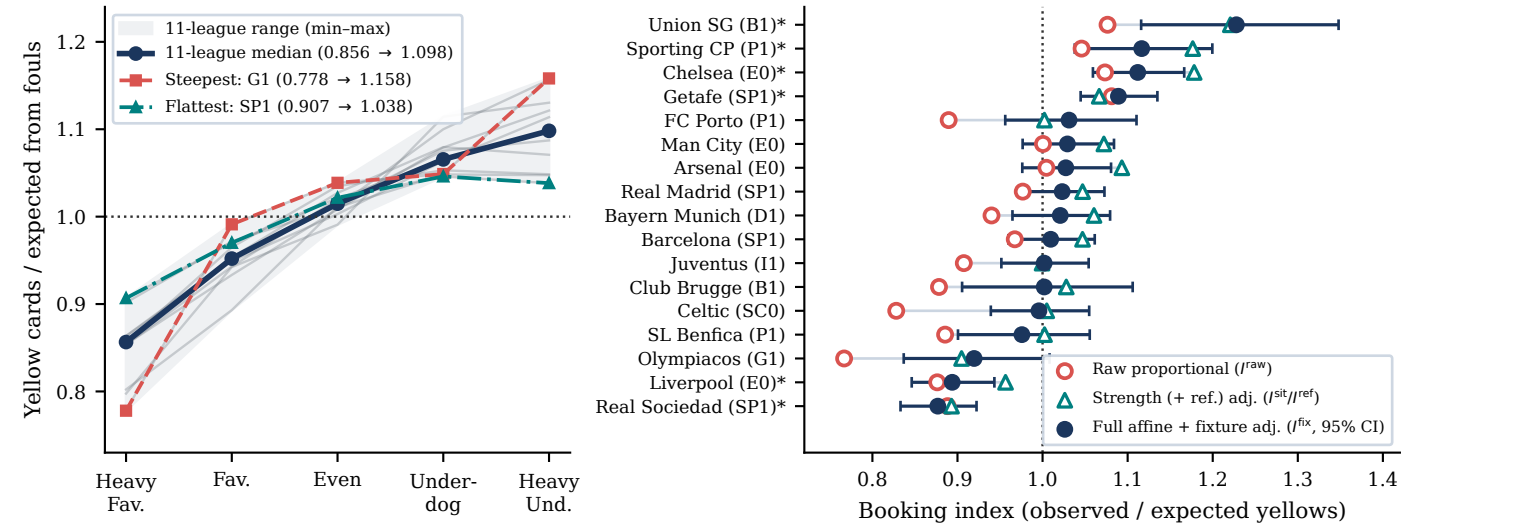
(a) Win-Expectation Gradient (11 Leagues, $N = 58,013$)
(b) Deconfounding Across European Champions & Outliers ($N = 246$)
11-league range (min–max)
11-league median (0.856 → 1.098)
Steepest: G1 (0.778 → 1.158)
Flattest: SP1 (0.907 → 1.038)
Yellow cards / expected from fouls
1.2
1.1
1.0
0.9
0.8
Heavy Fav.
Fav.
Even
Under-dog
Heavy Und.
Union SG (B1)*
Sporting CP (P1)*
Chelsea (E0)*
Getafe (SP1)*
FC Porto (P1)
Man City (E0)
Arsenal (E0)
Real Madrid (SP1)
Bayern Munich (D1)
Barcelona (SP1)
Juventus (I1)
Club Brugge (B1)
Celtic (SC0)
SL Benfica (P1)
Olympiacos (G1)
Liverpool (E0)*
Real Sociedad (SP1)*
Raw proportional (I^{raw})
Strength (+ ref.) adj. (I^{sit}/I^{ref})
Full affine + fixture adj. (I^{fix}, 95% CI)
0.8
0.9
1.0
1.1
1.2
1.3
1.4
Booking index (observed / expected yellows)